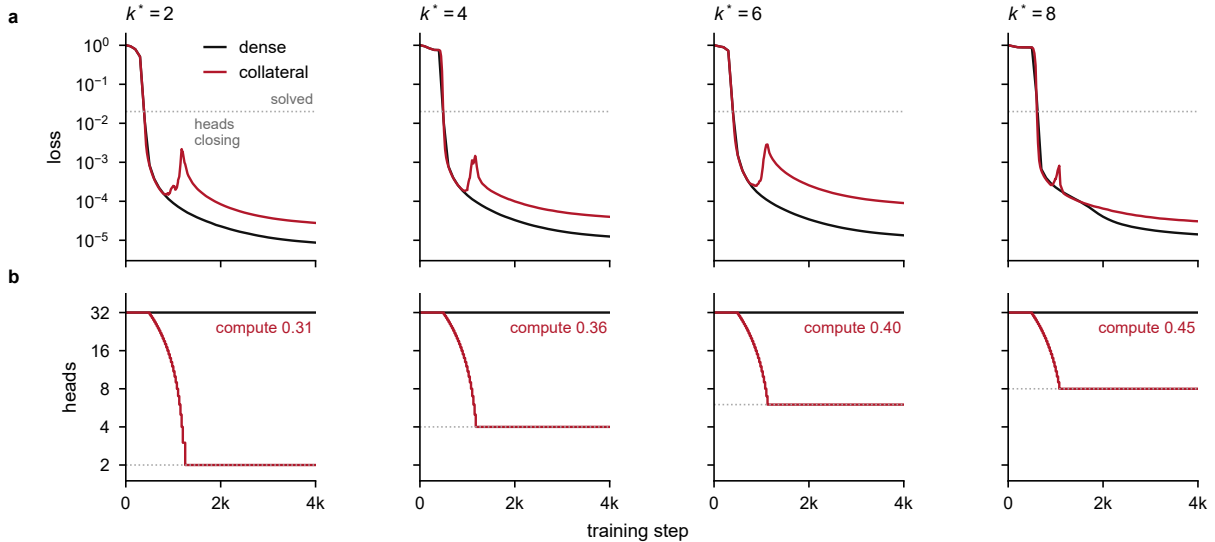




c Every method, same 30 runs

Method	Head score	During training	Gradual fade	Reopens	exact count never broke
Collateral rule (ours)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	•	•	•	29/30 30/30
<i>Published methods</i>					
ZipLM-style (Kurtic et al. 2023)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	—	—	—	24/30 0/30
Michel et al. 2019	$ \partial L / \partial g_h $ gradient of the head gate	•	—	—	28/30 0/30
Learned gates (Voita et al. 2019)	$L + \lambda \sum_h P(g_h \neq 0)$ a learned gate per head (L0 penalty)	•	—	—	9/30 0/30
<i>Controls: our rule with one part removed</i>					
no fade (instant closing)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	•	—	•	30/30 0/30
no re-fit	$L_h(W) - L(W)$ Optimal Brain Damage (LeCun et al. 1990)	•	•	•	2/30 30/30
no score (random choice)	random head	•	•	•	30/30 6/30

Tasks needing 2, 4 and 8 heads, 10 seeds each. • yes — no

